

GuardShield AI: Complete End-to-End Pipeline & Architecture Manual

Full Technical Implementation Guide: Data Ingestion, Model Training, Proxy Microservice & UI Deployment

Manual Overview: This document provides the complete, step-by-step technical execution blueprint for GuardShield AI.

STAGE 1: HIGH-LEVEL SYSTEM PIPELINE ARCHITECTURE

[1. RAW DATASETS] [2. FINE-TUNING PIPELINE] [3. INFERENCE PROXY] [JailbreakBench] [PyTorch + Hugging Face] [FastAPI Sidecar Proxy] [HackAPrompt] [DeBERTa-v3 Classifier] (Pre-Scan Guardrail) [deepset] (MPS / CUDA Acceleration) [(Clean Prompt)] [HaluEval] [NLI Entailment Classifier] [FEVER] + Token Logit Entropy [Local LLM (Ollama)] [TruthfulQA] (Mid-Sentence Inspector) [(Streaming Tokens)] [5. TELEMETRY DASHBOARD] [4. POST-SCAN FILTER] (Streamlit / React Analytics) (NLI + Entropy Verification)